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Method, information processing apparatus, and storage medium

Abstract

A method of collecting operation information from an apparatus on a network includes searching for an apparatus on the network, in response to finding an image processing apparatus and a control apparatus that is connected to the image processing apparatus and transmits print data to the image processing apparatus in the searching, collecting operation information from one of the image processing apparatus and the control apparatus, and storing the collected operation information in a predetermined storage area.

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Background/Summary

BACKGROUND

Field of the Disclosure

(1) The present disclosure relates to operation management of a system in which an information processing apparatus and a control apparatus that is connected to the image processing apparatus and transmits print data to the image processing apparatus are connected to the same network.

Description of the Related Art

(2) In recent years, most of image processing apparatuses such as printers, scanners, facsimiles (FAX), and multifunction printers (MFP) compositely including these functions have included a network communication function. A MFP including the communication function communicates with servers and personal computers (PC) on a network, to carry out functions such as scan data transmission from the MFP to the PCs.

(3) On the other hand, professional-use products for printing companies and design offices are required to perform more advanced print controls than products for general offices, for example, in terms of image processing and scheduling of print jobs. Such advanced print controls are performable with a control apparatus called a digital front end controller (DFE) connected to the MFP. Printing companies and design offices often uses a DFE connected to an MFP.

(4) Although a DFE and an MFP include device information in their own control units, there is

demand to collectively manage device information that is currently managed by the control units of the DFE and the MFP in a distributed way. Japanese Patent Application Laid-Open No. 2020-091370 discusses a technique in which a DFE connected to an MFP in a network acquires device information from the MFP in response to a request from a device management apparatus, and merges the acquired device information with own device information, thereby responding to the request. That technique allows device information managed by a DFE and an MFP in a distributed way to be collectively managed.

SUMMARY

(5) According to embodiments of the present disclosure, a method of collecting operation information from an apparatus on a network includes searching for an apparatus on the network, in response to finding an image processing apparatus and a control apparatus that is connected to the image processing apparatus and transmits print data to the image processing apparatus in the searching, collecting operation information from one of the image processing apparatus and the control apparatus, and storing the collected operation information in a predetermined storage area.

(6) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of a connection environment of a device operation management system according to a present exemplary embodiment.

(2) FIG. 2 is a block diagram illustrating an example of a hardware configuration of a multifunction printer (MFP) and a digital front end controller (DFE).

(3) FIG. 3 is a block diagram illustrating an example of a hardware configuration of an information collection personal computer (PC).

(4) FIG. 4 is a block diagram illustrating an example of a hardware configuration of a device operation management apparatus.

(5) FIG. 5 is a block diagram illustrating an example of a processing function configuration of the information collection PC.

(6) FIG. 6 is a block diagram illustrating an example of a processing function configuration of the device operation management apparatus.

(7) FIGS. 7A to 7C are diagrams illustrating data structures of a device list, transmission operation information, and operation information according to a first exemplary embodiment.

(8) FIG. 8 is a flowchart illustrating an example of processing performed by an information collection PC according to the first exemplary embodiment.

(9) FIG. 9 illustrates an example of a setting screen for an information collection PC according to a second exemplary embodiment.

(10) FIG. 10 is a flowchart illustrating an example of processing performed by the information collection PC according to the second exemplary embodiment.

(11) FIG. 11 illustrates an example of an operation state screen according to a third exemplary embodiment.

(12) FIG. 12 illustrates a display example of detailed information on a device according to the third exemplary embodiment.

(13) FIGS. 13A and 13B are diagrams illustrating data structures of the device list and the transmission operation information according to the third exemplary embodiment.

(14) FIG. 13C is a diagram illustrating a data structure of the operation information according to the third exemplary embodiment.

(15) FIG. 14A is a flowchart illustrating an example of processing performed by a device operation

management apparatus according to the third exemplary embodiment.

(16) FIG. 14B is a flowchart illustrating an example of processing performed by the device operation management apparatus according to the third exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

(17) There is a case where, with a digital front end controller (DFE) connected to a multifunction printer (MFP), both the MFP and the DFE are connected to a network. In that case, operation information is collected from both the MFP and the DFE, and is transmitted to a management apparatus. That leads to management of the MFP and the DFE separately by the management apparatus. In the present exemplary embodiment, a mechanism is provided that prevents one device whose operation state is to be managed from being viewed as if to be two devices from the user. More specifically, the mechanism allows the management apparatus in a system with an image processing apparatus and a control apparatus that is connected to the image processing apparatus and transmits print data both connected to a network to manage operation of the image processing apparatus without duplication.

(18) Some exemplary embodiments of the present disclosure will be specifically described below with reference to drawings. Configurations described in the following exemplary embodiments are illustrative, and the present disclosure is not limited to the illustrated configurations.

(19) <Configuration of Device Operation Management System>

(20) FIG. 1 illustrates an example of a connection environment of a device operation management system according to a first exemplary embodiment of the present disclosure.

(21) As illustrated in FIG. 1, the device operation management system (hereinafter, referred to as “the system”) according to the present exemplary embodiment includes an MFP **101** and a DFE **102**. The system further includes a client personal computer (PC) **103**, an information collection PC **104**, and a device operation management apparatus **105**.

(22) The MFP **101**, the DFE **102**, the client PC **103**, and the information collection PC **104** are communicably connected to one another through a network **106**.

(23) Further, the DFE **102** and the MFP **101** are connected via an internal interface **107**, which transfers images at high speed.

(24) In the system, the client PC **103** transmits a print job through the network **106** to the DFE **102**. The DFE **102** sequentially performs rasterizing processing on the print job received from the client PC **103** through the network **106**, and transfers rasterized image data (also referred to as print data) to the MFP **101** through the internal interface **107**. The MFP **101** performs suitable processing on the rasterized image data received from the DFE **102** through the internal interface **107**, and performs printing and output on an actual sheet.

(25) Print jobs for offices can be transmitted from the client PC **103** to the MFP **101** through the network **106**. The MFP **101** sequentially processes the print job received from the client PC **103**, and performs printing and output on an actual sheet by a marking engine **220** (FIG. 2). Further, the MFP **101** can perform copy (copy job) with a connected scanner, and transmit a scan result to the client PC **103** (transmission job). The MFP **101** can print facsimile (FAX) reception data (FAX job) received through a telephone line (not illustrated).

(26) The DFE **102** communicates with the MFP **101**, and acquires and manages the serial number, product name information, etc. of the MFP **101**. Further, the DFE **102** manages execution information on a job carried out by itself.

(27) The information collection PC **104** is an information processing apparatus that searches for a device connected to the network **106** and registers a found device in a device list described below. The information collection PC **104** collects information on devices registered in the device list, through the network **106**, and stores the collected information. Further, the information collection PC **104** transmits the collected and stored information to the device operation management apparatus **105**.

(28) The device operation management apparatus **105** displays information that is browsed by a

print administrator in order to manage productivity and an operation rate of a plurality of MFPs. The device operation management apparatus **105** includes units to receive information from the information collection PC **104**, to store the information, and to prepare display of a device operation state. The device operation management apparatus **105** may consist of one computer or a plurality of computers. The device operation management apparatus **105** may be provided in, for example, a cloud service.

(29) The client PC **103** accesses the device operation management apparatus **105**, and displays the information prepared for display by the device operation management apparatus **105**. This enables the print administrator to check the operation state of a device of interest. The client PC **103** is any of various kinds of information processing apparatuses such as a personal computer, a tablet terminal, and a smartphone.

(30) <Device Configuration of MFP **101** and DFE **102**>

(31) FIG. **2** is a block diagram illustrating an example of a hardware configuration of the MFP **101** and the DFE **102**.

(32) The MFP **101** includes a controller unit including a central processing unit (CPU) **211**, a random access memory (RAM) **212**, and a hard disk drive (HDD) **213**. The MFP **101** further includes the marking engine **220**, a scanner **221**, an image processing unit **216**, an image processing interface (I/F) unit **217**, a device I/F unit **218**, and a network I/F unit **214**. Various kinds of programs are stored in the HDD **213**, are read out to the RAM **212**, and are run by the CPU **211** as appropriate.

(33) The DFE **102** includes a controller unit including a CPU **201**, a RAM **202**, and an HDD **203**. The DFE **102** further includes an image processing unit **205**, an image processing I/F unit **206**, and a network I/F unit **204**. Various kinds of programs are stored in the HDD **203**, are read out to the RAM **202**, and are run by the CPU **201** as appropriate. The DFE **102** functions as a control apparatus providing functions to the MFP **101**.

(34) The system with the MFP **101** and the DFE **102** connected to each other as described above is regarded as a system with a plurality of control units connected to an operation apparatus.

(35) In some embodiments, the MFP **101** and the DFE **102** each include a display/operation device (not illustrated) as appropriate.

(36) <Device Configuration of Information Collection PC **104** and Device Operation Management Apparatus **105**>

(37) FIG. **3** is a block diagram illustrating an example of a hardware configuration of the information collection PC.

(38) The information collection PC **104** includes a CPU **301**, a RAM **302**, an HDD **303**, and a network I/F unit **304**. Pieces of hardware such as the CPU **301**, the RAM **302**, and the HDD **303** constitute a so-called computer. Various kinds of programs are stored in the HDD **303**, are read out to the RAM **302**, and are controlled and run by the CPU **301** as appropriate.

(39) FIG. **4** is a block diagram illustrating an example of a hardware configuration of the device operation management apparatus **105**.

(40) The device operation management apparatus **105** includes a CPU **401**, a RAM **402**, an HDD **403**, and a network I/F unit **404**. Pieces of hardware such as the CPU **401**, the RAM **402**, and the HDD **403** constitute a so-called computer. Various kinds of programs are stored in the HDD **403** and the like, are read out to the RAM **402**, and are controlled and run by the CPU **401** as appropriate.

(41) <Processing Function Configuration of Information Collection PC **104**>

(42) FIG. **5** is a block diagram illustrating an example of a processing function configuration of the information collection PC **104**.

(43) Software modules of the information collection PC **104** illustrated in FIG. **5** are stored in the HDD **303** of the information collection PC **104**, and function by being read out to the RAM **302** and controlled and run by the CPU **301** as appropriate. In the information collection PC **104**, one

CPU **302** performs processing illustrated in a flowchart described below; however, other modes may be adopted. For example, the information collection PC **104** can cause a plurality of processors and memories to cooperate with one another to perform the processing illustrated in the flowchart described below. Further, part of processing such as transmission and reception of data is performed in cooperation with a hardware circuit such as a communication interface.

(44) A network management unit **501** uses the network I/F **304** to communicate with devices connected to the network **106**, for example, the DFE **102**, the MFP **101**, and the client PC **103**, and the device operation management apparatus **105** on the cloud, through the network.

(45) A device search processing unit **502** is a software module searching for a device connected to the network **106**. The device search processing unit **502** is run, for example, periodically, at the time of new setting, or at the time of adding a new device, and registers an operation management target device on the network in a device list described below.

(46) A device operation information collection processing unit **503** is a software module collecting operation information from the operation management target device registered in the device list. The collected operation information is stored in a data storage unit **505**. The data storage unit **505** stores the information in the HDD **303** of the information collection PC **104**. In some embodiments, the data storage unit **505** stores the information in a predetermined storage area such as the RAM **302** or an external storage area.

(47) A device operation information transmission processing unit **504** is a software module periodically transmitting device operation information that is collected by the device operation information collection processing unit **503** and is stored in the data storage unit **505**, to the device operation management apparatus **105**.

(48) <Processing Function Configuration of Device Operation Management Apparatus **105**>

(49) FIG. **6** is a block diagram illustrating an example of a processing function configuration of the device operation management apparatus **105**.

(50) Software modules of the device operation management apparatus **105** illustrated in FIG. **6** are stored in the HDD **403** of the device operation management apparatus **105**, and function by being read out to the RAM **402** and controlled and run by the CPU **401** as appropriate. In the device operation management apparatus **105**, one CPU **401** performs processing illustrated in a flowchart described below; however, other modes may be adopted. For example, the device operation management apparatus **105** can cause a plurality of processors and memories to cooperate with one another to perform the processing in the flowchart described below. Further, part of processing such as transmission and reception of data is performed in cooperation with a hardware circuit such as a communication interface.

(51) A network management unit **601** uses the network I/F unit **104** to communicate with the information collection PC **104**, the client PC **103**, and a device on the Internet.

(52) An operation information reception processing unit **602** is a software module receiving operation information transmitted from the information collection PC **104**.

(53) A device operation information management unit **603** is a software module that appropriately manages device operation information received by the operation information reception processing unit **602** and stores the device operation information in a data storage unit **605**. The data storage unit **605** stores the information in the HDD **403** of the device operation management apparatus **105**. In some embodiments, the data storage unit **605** stores the information in a predetermined storage area such as the RAM **402** or an external storage area.

(54) A device operation information display processing unit **604** is a software module displaying device operation information managed by the device operation information management unit **603**, on the client PC **103** or the device on the Internet. The device operation information display processing unit **604** generates information representing a device operation state (prepares display screen representing device operation state) based on device operation information managed by the device operation information management unit **603**.

(55) FIG. 7A is a diagram illustrating an example of a data structure of a device list managed by the device operation information management unit **603** according to the first exemplary embodiment.

(56) The device list includes information on a plurality of devices to be managed (hereinafter, “devices **702**”). Each of the devices **702** has device information **703** and MFP information **705**. The device information **703** includes information **704** such as the product name and the serial number of the corresponding device. The MFP information **705** includes information **706** such as the internet protocol (IP) address and the media access control (MAC) address of the corresponding MFP.

(57) FIG. 7B is a diagram illustrating an example of a data structure of transmission operation information **711** transmitted from the MFP **101** to the information collection PC **104**.

(58) The transmission operation information **711** includes, for each job, information **712** such as a serial number, a date, a product name, counter information, a job type, detail of the job type, information on the number of jobs after previous transmission, and information on the number of printed sheets.

(59) FIG. 7C is a diagram illustrating an example of a data structure of operation information managed by the device operation information management unit **603** according to the first exemplary embodiment.

(60) Operation information **721** is managed for each device serial number **722**, and includes the total number of jobs **724** and the total number of pages **725** for each job type **723**. Examples of the job type **723** include print job, copy job, facsimile job, and scan job.

(61) FIG. 8 is a flowchart illustrating an example of processing performed by the device search processing unit **502** in the information collection PC **104** according to the first exemplary embodiment, to search for a device on the network and to register the found device in a device list. The CPU **301** of the information collection PC **104** loads programs stored in the HDD **303** or other storages to the RAM **302** and run the programs as appropriate to perform the processing of the flowchart.

(62) In step **S801**, the device search processing unit **502** searches for a device connected to the network I/F unit **304** through the network **106**. For example, the device search processing unit **502** broadcasts search packets. In place of broadcasting, multicasting may be performed.

(63) Next, in step **S802**, the device search processing unit **502** acquires device information on a device that has responded to in the search in step **S801**, and stores the device information in the memory **302** or the HDD **303** through the data storage unit **505**.

(64) Next, in step **S803**, the device search processing unit **502** determines whether the device information has been registered in the device list. If the device information has been registered (YES in step **S803**), the processing of the flowchart ends.

(65) Otherwise (NO in step **S803**), the processing proceeds to step **S804**.

(66) In step **S804**, the device search processing unit **502** determines whether the device information acquired in step **S802** is device information on the MFP. If the device information acquired in step **S802** is not device information on the MFP (NO in step **S804**), the processing of the flowchart ends without adding the device information acquired in step **S802** to the device list.

(67) If it is determined that the device information acquired in step **S802** is device information on the MFP (YES in step **S804**), the processing proceeds to step **S805**.

(68) In step **S805**, the device search processing unit **502** adds the device information acquired in step **S802** to the device list. More specifically, the device search processing unit **502** adds the pieces of information **702** to **706** to the device list **701**, which has the structure as illustrated in FIG. 7A, based on the device information acquired in step **S802**. After the processing in step **S805**, the processing of the flowchart ends.

(69) The device list registered in the above-described manner is stored in the data storage unit **505** and is transmitted to the device operation management apparatus **105**, and is managed by the device operation information management unit **603** of the device operation management apparatus

105. Further, the device operation information collection processing unit **503** of the information collection PC **104** collects the operation information as illustrated in FIG. 7B from the device registered in the above-described device list, and stores the collected operation information in the data storage unit **505**. Further, the device operation information transmission processing unit **504** transmits the operation information stored in the data storage unit **505** to the device operation management apparatus **105**. The device operation information management unit **603** of the device operation management apparatus **105** manages the operation information on the MFP **101** (including serial number of MFP **101**) as illustrated in FIG. 7B received from the information collection PC **104**, with the data structure as illustrated in FIG. 7C, based on the serial number.

(70) According to the first exemplary embodiment, in the print system with the MFP and the DFE both connected to a network, the above-described mechanism allows the device operation management apparatus to manage the operation state without duplication of the same MFP. This prevents some pieces of operation information on a single MFP from being displayed as if to be respective pieces of operation information on different MFPs.

(71) A second exemplary embodiment will be described. In the above-described first exemplary embodiment, the configuration has been described that registers the found MFP in the device list without registering the DFE connected to the MFP in the device list to manage the operation information without duplication. Some users have demand to manage operation information on the DFE in detail. The following is a description of an exemplary embodiment satisfying the demand.

(72) FIG. 9 illustrates an example of a setting screen for the information collection PC **104** according to the second exemplary embodiment.

(73) The screen is displayed on a display unit of the client PC **103** in response to access of the client PC **103** to the information collection PC **104**. A user such as a print administrator performs instructions on the screen.

(74) A setting screen **901** is a screen that allows the user to previously select and set an acquisition source of device operation information. The user selects an acquisition source of device operation information with a radio button **902** or **903**. To acquire device operation information from the MFP **101**, the user selects the radio button **902**. Otherwise, to acquire device operation information from the DFE **102**, the user selects the radio button **903**. A selection of one of the radio buttons **902** and **903** and then a press of an “OK” button **904** enables the setting in the setting screen **901**, and then the setting screen **901** is closed. The setting is stored in the HDD **303** of the information collection PC **104**. A press of a “cancel” button **905** disables the setting in the setting screen **901**, and the setting screen **901** is closed.

(75) In the second exemplary embodiment, the “MFP information” **705** in FIG. 7A is replaced with “device information acquisition source information”. In addition, if the DFE **102** is set as the device operation information acquisition source in the setting screen **901**, transmission operation information transmitted from the DFE **102** to the information collection PC **104** includes the serial number of the MFP **101**, which is acquired from the MFP **101** and is managed by the DFE **102**.

(76) FIG. 10 is a flowchart illustrating an example of processing performed by the device search processing unit **502** in the information collection PC **104** according to the second exemplary embodiment, to search for a device on the network and to register the found device in the device list. The CPU **301** of the information collection PC **104** loads programs stored in the HDD **303** or other storages to the RAM **302** and runs the programs as appropriate to perform the processing in the flowchart.

(77) The processing in steps **S1001** to **S1003** is the same as the processing in steps **S801** to **S803** in FIG. 8. Thus, the description of the processing will be omitted.

(78) In step **S1004**, the device search processing unit **502** determines whether the device information acquired in step **S1002** is device information on the MFP or device information on the DFE, and determines whether the device corresponding to the device information acquired in step **S1002** is the device operation information acquisition source set in the screen **901** of FIG. 9. If the

device corresponding to the device information acquired in step **S1002** is not the device operation information acquisition source (NO in step **S1004**), the processing of the flowchart ends without adding the device information acquired in step **S1002** in the device list.

(79) On the other hand, if it is determined that the device corresponding to the device information acquired in step **S1002** is the device operation information acquisition source (YES in step **S1004**), the processing proceeds to step **S1005**.

(80) In step **S1005**, the device search processing unit **502** adds the device information acquired in step **S1002** in the device list. More specifically, the device search processing unit **502** adds the pieces of information **702** to **706** to the device list **701** having the above-described structure, based on the device information acquired in step **S1002**. After the processing in step **S1005**, the processing of the flowchart ends.

(81) The device list registered in the above-described manner is stored in the data storage unit **505** and is transmitted to the device operation management apparatus **105**, and is managed by the device operation information management unit **603** of the device operation management apparatus **105**. Further, the device operation information collection processing unit **503** of the information collection PC **104** collects operation information from the device registered in the above-described device list, and the device operation information transmission processing unit **504** transmits the collected operation information to the device operation management apparatus **105**. The device operation information management unit **603** of the device operation management apparatus **105** manages operation information on the MFP **101** or the DFE **102** (including serial number of MFP **101**) received from the information collection PC **104**, with the data structure as illustrated in FIG. 7C, based on the serial number.

(82) The described-above mechanism according to the second exemplary embodiment allows operation information to be managed without duplication of the same MFP, by using operation information acquired from the device selected and set as the device information acquisition source. A selection and set of a DFE as a device information acquisition source by a print administrator allows detailed management of operation information on the DFE.

(83) A third exemplary embodiment will be described. In the above-described first and second exemplary embodiments, if a device on the network is searched for and the MFP and the DFE connected to the MFP are found, operation information is collected from one of the MFP and the DFE. Further, the device operation management apparatus **105** manages device operation information using the information collected in the above-described manner. In other words, the configuration has been described that manages the operation information without duplication of the same MFP using the operation information collected from one of the MFP and the DFE. On the other hand, the DFE and the MFP run jobs different from each other. As a result, users who want to analyze the number of printed sheets and the number of jobs in detail have demand to integrally manage operation information on the MFP and operation information on the DFE. In the present exemplary embodiment, a configuration will be described that integrally manages operation information on the MFP and operation information on the DFE. In the present exemplary embodiment, the information collection PC **104** collects operation information from both the MFP **101** and the DFE **102**.

(84) FIG. 11 illustrates an example of an operation state screen displaying, on the client PC **103**, display content prepared by the device operation information display processing unit **604** according to the third exemplary embodiment.

(85) In an operation state screen **1101**, a device list is displayed in a device list **1102** on the left of the screen. The MFP **101** and the DFE **102** are recognized as two different devices on the network, but the information is displayed as a single device in the device list. In the example in FIG. 11, a device “AB-Cx00” displayed at the head of the device list **1102** is a device including the MFP **101** and the DFE **102**, which is displayed as a single device. The MFP **101** and the DFE **102** are connected via the internal interface **107**, and each transmit transmission operation information

including the same serial number to the information collection PC **104**.

(86) A consumable information display displays the remaining amounts of consumables.

(87) When any of the devices is selected from the device list displayed in the device list **1102** and an instruction is issued (e.g., double click is performed), for example, detailed information **1204** in FIG. **12** on the selected device is displayed.

(88) FIG. **12** illustrates a display example of the detailed information on the device according to the third exemplary embodiment.

(89) In the detailed information **1204**, the total number of print jobs and the total number of pages are displayed as combined (integrated) information on the jobs run by both the MFP **101** and the DFE **102** constituting the device selected in the device list **1102**.

(90) FIGS. **13A** to **13C** illustrate examples of data structures of the device list, transmission operation information, and operation information according to the third exemplary embodiment.

(91) FIG. **13A** illustrates an example of the data structure of the device list managed by the device operation information management unit **603**.

(92) A device list **1301** includes information on a plurality of devices (hereinafter, “devices **1302**”).

(93) Each of the devices **1302** has device information **1303**, MFP information **1305**, and DFE information **1307**.

(94) The device information **1303** includes device information **1304** common to the MFP and the DFE, for example, a serial number and an engine name acquired from the MFP or the DFE. As counter information, more accurate information acquired from the MFP **101** may be used. The MFP information **1305** includes MFP unique information **1306** such as the IP address and the MAC address of the corresponding MFP. Likewise, the DFE information **1307** includes DFE unique information **1308** such as the DFE product name, the IP address, and the MAC address of the corresponding DFE.

(95) The device search processing unit **502** according to the third exemplary embodiment searches for a device on the network **106** (similar to step **S801** in FIG. **8**), acquires device information on the device that has responded to (similar to step **S802**), and stores the acquired device information in the memory **302** or the HDD **303** through the data storage unit **505**. Further, if the device information has not been registered in the device list (similar to step **S803**), the device search processing unit **502** adds the information **1302** and subsequent information in the device list **1301** having the structure as illustrated in FIG. **13A**, based on the acquired device information. At this time, if the acquired device information is device information on the MFP, the pieces of MFP information **1305** and **1306** are added. If the acquired device information is device information on the DFE, the pieces of DFE information **1307** and **1308** are added.

(96) The device list registered in the above-described manner is stored in the data storage unit **505** and is transmitted to the device operation management apparatus **105**, and is managed by the device operation information management unit **603** of the device operation management apparatus **105**. Further, the device operation information collection processing unit **503** of the information collection PC **104** collects operation information from the device registered in the above-described device list, and the device operation information transmission processing unit **504** transmits the collected operation information to the device operation management apparatus **105**.

(97) FIG. **13B** illustrates an example of data structure of transmission operation information **1311** transmitted from the MFP **101** or the DFE **102** to the information collection PC **104**.

(98) The transmission operation information **1311** includes, for each job, information **1312** such as a serial number, a date, a product name, counter information, a job type, detail of the job type, and information on the number of printed sheets. The serial number included in the transmission operation information transmitted from the DFE **102** to the information collection PC **104** corresponds to the serial number of the MFP **101** that is acquired from the MFP **101** and managed by the DFE **102**.

(99) FIG. **13C** illustrates an example of a data structure of operation information managed by the

device operation information management unit **603**.

(100) Operation information **1321** is managed for each of devices (hereinafter, “devices **1322**”). Each of the devices **1322** has device information **1323**, MFP operation information **1325**, and DFE operation information **1336**.

(101) In the device information **1323**, information **1324** such as a serial number, counter information, and consumable information is managed.

(102) The MFP operation information **1325** is based on operation information collected from the MFP. The MFP operation information **1325** includes the total number of jobs and the total number of pages in each of job types such as a print job **1326**, a copy job **1333**, a scan job **1334**, and a FAX job **1335**.

(103) The print job **1326** is specifically classified into a DFE job **1327** and an MFP job **1328**. The DFE job **1327** is print job information input from the DFE **102** to the MFP **101**. The DFE job **1327** includes the total number of jobs **1329** and the total number of pages **1330**. Likewise, the MFP job **1328** includes the total number of jobs **1331** and the total number of pages **1332**.

(104) The DFE operation information **1336** is based on operation information collected from the DFE. The DFE operation information **1336** includes a print job **1337** and a scan job **1340**.

(105) The print job **1337** includes a portable document format (PDF) job **1338** and a PostScript job **1339** as detailed print types, and the total number of jobs and the total number of pages are managed for each job.

(106) In the scan job **1334**, the total number of jobs and the total number of pages are managed for each job as combined information on the scan job run by the MFP and the scan job run in response to an instruction from the DFE.

(107) The DFE job **1327** and the print job **1337** each include at least information on update date and time (not illustrated).

(108) The device operation information management unit **603** of the device operation management apparatus **105** associates operation information on the MFP **101** and operation information on the DFE **102** (including serial number of MFP) as illustrated in FIG. **13B** received from the information collection PC **104**, with a serial number, and manages the operation information with the data structure as illustrated in FIG. **13C**. At this time, it is assumed that the serial number included in the operation information collected from the MFP **101** is the serial number included in the operation information collected from the DFE **102**. In other words, the operation information on the MFP **101** and the operation information on the DFE **102** are managed in association with each other with the serial number as a key.

(109) FIG. **14A** and FIG. **14B** are flowcharts illustrating an example of processing performed by the device operation information display processing unit **604** in the device operation management apparatus **105** according to the third exemplary embodiment, to create operation information to be displayed on the client PC **103**. The CPU **401** of the device operation management apparatus **105** loads programs stored in the HDD **403** or other storages to the RAM **402** and runs the programs as appropriate to perform processing illustrated in FIG. **14A** and FIG. **14B**.

(110) First, in step **S1401**, the device operation information display processing unit **604** requests the device operation information management unit **603** to acquire operation information (MFP operation information **1325** and DFE operation information **1336**) on one of the devices in the device list. In the following, the device, the operation information on which is acquired, is referred to as a “corresponding device”.

(111) In step **S1402**, the device operation information display processing unit **604** checks whether the last operation information in the device list has been acquired. If the last operation information in the device list has not been acquired, namely, if the device, no operation information on which has been acquired, is present in the device list (NO in step **S1402**), the processing proceeds to step **S1403**. In this case, the operation information on the device requested in step **S1401** is acquired.

(112) In steps **S1403** to **S1405**, the device operation information display processing unit **604**

combines the MFP operation information **1325** and the DFE operation information **1336** acquired in step **S1401**, to create display information. Each of the steps will be described.

(113) In step **S1403**, the device operation information display processing unit **604** creates print job operation display information on the corresponding device with the operation information acquired in the step **S1401**. The detail of the processing will be described with reference to FIG. **14B**.

(114) Next, in step **S1404**, the device operation information display processing unit **604** creates scan job operation display information on the corresponding device with the operation information acquired in step **S1401**.

(115) Next, in step **S1405**, the device operation information display processing unit **604** creates operation display information on other jobs such as the FAX job **1335** on the corresponding device with the operation information acquired in step **S1401**.

(116) Next, in step **S1406**, the device operation information display processing unit **604** requests the device operation information management unit **603** to acquire operation information on the next device from the device list. Thereafter, the processing proceeds to step **S1402**.

(117) If the last operation information in the device list has been already acquired, namely, if no information on the device, operation information on which is to be acquired, is present in the device list (YES in step **S1402**), the processing of the flowchart ends.

(118) The operation display information created (prepared for display) by the device operation management apparatus **105** in the above-described manner is transmitted to the client PC **103** in response to a request from the client PC **103**, and is displayed on the client PC **103**.

(119) FIG. **14B** is a flowchart illustrating details of the processing to create the print job operation display information in step **S1403** of FIG. **14A**.

(120) First, in step **S1411**, the device operation information display processing unit **604** checks whether the DFE job **1327** in the MFP operation information **1325** matches the print job **1337** in the DFE operation information **1336**, through comparison. If the DFE job **1327** in the MFP operation information **1325** and the print job **1337** in the DFE operation information **1336** are equal in the total number of jobs and the total number of pages, it is determined that the DFE job **1327** in the MFP operation information **1325** matches the print job **1337** in the DFE operation information **1336**.

(121) If the DFE job **1327** in the MFP operation information **1325** matches the print job **1337** in the DFE operation information **1336** (YES in step **S1411**), the processing proceeds to step **S1414**.

(122) In step **S1414**, the device operation information display processing unit **604** uses the information on the print job **1337** in the DFE operation information **1336**. In this case, in next step **S1415**, the device operation information display processing unit **604** combines the MFP job **1328** in the MFP operation information **1325** and the print job **1337** in the DFE operation information **1336**, to acquire print job total operation information on the device. Performing the processing makes it possible to integrate the result of the print job run by the single MFP with print results managed by the DFE without going through the DFE. For example, results of status print to print status information on the MFP and results of print jobs that have been directly received by the MFP from mobile terminals and performed can be correctly counted.

(123) In contrast, if the DFE job **1327** in the MFP operation information **1325** does not match the print job **1337** in the DFE operation information **1336** (NO in step **S1411**), the processing proceeds to step **S1412**.

(124) In step **S1412**, the device operation information display processing unit **604** compares the DFE job **1327** in the MFP operation information **125** and the print job **1337** in the DFE operation information **1336** in the time when the operation information was received from the device to determine which is the latest information. If the print job **1337** in the DFE operation information **1336** is newer (DFE in step **S1412**), the processing proceeds to step **S1414**.

(125) In contrast, if the DFE job **1327** in the MFP operation information **1325** is newer (MFP in step **S1412**), the processing proceeds to step **S1413**.

(126) In step **S1413**, the device operation information display processing unit **604** uses the information on the DFE job **1327** in the MFP operation information **1325**. In this case, in next step **S1415**, the device operation information display processing unit **604** combines the MFP job **1328** in the MFP operation information **1325** and the DFE job **1327** in the MFP operation information **1325**, to acquire print job total operation information on the device.

(127) Next, in step **S1416**, the device operation information display processing unit **604** acquires detailed job information (information on PDF job **1338** and PostScript job **1339**) in the print job **1337** in the DFE operation information **1336**.

(128) Next, in step **S1417**, the device operation information display processing unit **604** calculates the ratio of each job based on the print job total operation information acquired in step **S1415** and the detailed job information acquired in step **S1416**, and creates information for display. At this time, a display screen is created so that the ratio of each job is displayed in descending order of the number of jobs. The detail display screen is created to add the ratio of the PDF job **1338** and the ratio of the PostScript job **1339**. Further, the display screen is created to display the ratio of a job through the DFE and the ratio of a job processed by the MFP without going through the DFE, in the same job type.

(129) After the processing in step **S1417**, the processing of the flowchart ends. As described above, the information duplicated between the operation information on the MFP **101** and the operation information on the DFE **102** is combined (integrated) using one of the pieces of operation information on the MFP **101** and on the DFE **102**.

(130) As described above, in the present exemplary embodiment, when the executed job type is displayed, the pieces of operation information on both the MFP and the DFE are combined and displayed. At this time, the number of copy jobs and the number of FAX jobs run by the MFP are acquired from the MFP operation information. In contrast, print jobs run by the DFE are acquired from the DFE operation information.

(131) According to the third exemplary embodiment, in the print system with the MFP and the DFE both connected to a network, the above-mentioned mechanism allows the device operation management apparatus to manage operation information without duplication of the same MFP, and to appropriately and integrally manage distributed job information. This makes it possible to prevent operation information on a single MFP from being displayed as if to be respective pieces of operation information on different MFPs.

(132) Since the DFE and the MFP run different jobs, detail analysis of the number of printed sheets and the number of jobs involves both the MFP operation information and the DFE operation information. In the present exemplary embodiment, both pieces of operation information are combined to create display information. This enables detailed analysis of the number of printed sheets and the number of jobs.

(133) In the above-described example, the configuration has been described that, if information on the same job type is included in both the MFP operation information and the DFE operation information, like the print job, both pieces of the information are combined to acquire the total information on the job type. In some embodiments, whether to combine both pieces of the information on the same job type in the MFP operation information and in the DFE operation information to acquire the total information is switched by user setting.

(134) Further, if information on the same page description language (PDL) type is included in both the MFP operation information and the DFE operation information, both pieces of information may be combined to acquire total information. Further, a display screen is created to display PDL types in descending order of the number of jobs. Furthermore, if detailed display is instructed, the ratio of jobs of each PDL type may be displayed.

(135) The display may include the ratio of jobs through the DFE and the ratio of job run by the MFP without going through the DFE in a PDL type. Additionally, whether to combine both pieces of information on the same PDL type in MFP operation information and in DFE operation

information may be switched by user setting.

(136) The device operation information display processing unit **604** may include a plurality of applications, and whether to perform combination or operation information used for display may be different depending on the application. For example, a first application displays and controls an operation state based on a result of combining MFP operation information and DFE operation information. A second application displays and controls an operation state based on MFP operation information to manage a job run by the MFP without going through the DFE. Further, a third application displays and controls an operation information based on information on the DFE for management in units of a print job. The client PC **103** can display a desired operation state by accessing a uniform resource locator (URL) of the application of which the operation state is to be displayed.

(137) In the above-described exemplary embodiments, the information collection PC **104** performs generation of a device list and collection and transmission of operation information. Alternatively, the MFP **101** or the DFE **102** may perform similar control. In other words, the MFP **101** or the DFE **102** may include the above-described functions of the information collection PC **104**.

(138) Further, collection control of operation information provided by the PC **104**, integral management control of operation information, creation of a screen about the operation information, and provision of the screen to the client PC provided by the management apparatus **105** may be performed by one server or one cloud service.

(139) In the print system with the MFP **101** and the DFE **102** both connected to a network, the mechanism allows the device operation management apparatus **105** to manage operation without duplication of the same MFP, and to integrally manage distributed job information.

(140) In other words, the printing apparatus is not displayed with duplication, the states of jobs run by the MFP and the DFE separately can be displayed in combination, and additionally a network communication fee can be reduced. As a result, in the system with a plurality of control units all connected to an operation apparatus, the operation state of the operation apparatus can be appropriately managed.

(141) According to the exemplary embodiments described above, in the system with the image processing apparatus and the control apparatus that is connected to the image processing apparatus and transmits print data to the image processing apparatus are both connected to a network, the management apparatus can manage operation without duplication of the image processing apparatus. This makes it possible to prevent pieces of operation information on a single MFP from being displayed as if to be respective pieces of operation information on different MFPs.

(142) It goes without saying that the configurations and the details of the various kinds of data described above are not limited thereto, and various configurations and details are used depending on applications and purposes.

(143) Although some exemplary embodiments are described above, embodiments of the present disclosure can be embodied as, for example, a system, an apparatus, a method, a program, or a storage medium. More specifically, embodiments of the present disclosure may be applied to a system including a plurality of devices or to an apparatus including one device.

(144) Further, embodiments of the present disclosure include combinations of the above-described exemplary embodiments.

OTHER EXEMPLARY EMBODIMENTS

(145) Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for

example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(146) While the present disclosure includes exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(147) This application claims the benefit of Japanese Patent Application No. 2021-065121, filed Apr. 7, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A method for an information processing apparatus, which communicates with an image processing apparatus and a control apparatus connected to the image processing apparatus and transmitting print data to the image processing apparatus, the method comprising: acquiring first operation information and second operation information from a data storage system, wherein the first operation information is based on first transmitted operation information from the image processing apparatus via a network, and wherein the second operation information is based on second transmitted operation information from the control apparatus via the network; and performing control to display device operation state information about one apparatus based on a total amount of one of first information that is information about a number of jobs through the control apparatus and included in the first operation information based on the first transmitted operation information including a specific identifier and second information that is information about a number of jobs through the control apparatus and included in the second operation information based on the second transmitted operation information including the specific identifier, and third information that is information about a number of jobs processed by the image processing apparatus without going through the control apparatus and included in the first operation information.
2. The method according to claim 1, further comprising using, in a case where a specific condition is satisfied, the second information without using the first information to display the device operating status information.
3. The method according to claim 2, further comprising, in a case where the specific condition is not satisfied, using the newer of the first information and the second information without using the older of the first information and the second information to display the device operating status information.
4. The method according to claim 1, further comprising outputting display data for displaying an operation state of the one apparatus.
5. The method according to claim 4, further comprising outputting the display data in response to a request from a client terminal, wherein the client terminal displays a screen representing the device operation state information based on the output display data.
6. The method according to claim 1, wherein the identifier is a serial number of the image processing apparatus.

7. The method according to claim 1, wherein the first transmitted operation information includes date and time, a product name, counter information, a job type, job type details, information on the number of jobs since a previous transmission, and information on the number of printed sheets.
8. The method according to claim 1, wherein the control apparatus is a Digital Front End (DFE) controller.
9. The method according to claim 1, further comprising managing the first transmitted operation information and the second transmitted operation information by serial number of each apparatus.
10. The method according to claim 1, wherein the control apparatus performs rasterizing processing on a received print job and transmits image data that has been rasterized to the image processing apparatus.
11. The method according to claim 1, further comprising transmitting a print job to the image processing apparatus or the control apparatus.
12. An information processing apparatus configured to communicate with an image processing apparatus and a control apparatus connected to the image processing apparatus and transmitting print data to the image processing apparatus, the information processing apparatus comprising: one or more processors; and at least one memory storing executable instructions, which when executed by the one or more processors, cause the information processing apparatus to perform operations comprising: acquiring first operation information and second operation information from a data storage system, wherein the first operation information is based on first transmitted operation information from the image processing apparatus via a network, and wherein the second operation information is based on second transmitted operation information from the control apparatus via the network; and performing control to display device operation state information about one apparatus based on a total amount of one of first information that is information about a number of jobs through the control apparatus and included in the first operation information based on the first transmitted operation information including a specific identifier and second information that is information about a number of jobs through the control apparatus and included in the second operation information based on the second transmitted operation information including the specific identifier, and third information that is information about a number of jobs processed by the image processing apparatus without going through the control apparatus and included in the first operation information.
13. A non-transitory computer-readable storage medium storing executable instructions, which when executed by one or more processors of an information processing apparatus configured to communicate with an image processing apparatus and a control apparatus connected to the image processing apparatus and transmitting print data to the image processing apparatus, cause the information processing apparatus to perform operations comprising: acquiring first operation information and second operation information from a data storage system, wherein the first operation information is based on first transmitted operation information from the image processing apparatus via a network, and wherein the second operation information is based on second transmitted operation information from the control apparatus via the network; and performing control to display device operation state information about one apparatus based on a total amount of one of first information that is information about a number of jobs through the control apparatus and included in the first operation information based on the first transmitted operation information including a specific identifier and second information that is information about a number of jobs through the control apparatus and included in the second operation information based on the second transmitted operation information including the specific identifier, and third information that is information about a number of jobs processed by the image processing apparatus without going through the control apparatus and included in the first operation information.
14. The method according to claim 2, wherein the specific condition is that the first information matches the second information.

15. The method according to claim 2, wherein the specific condition is that a total number of jobs and a total number of pages that are indicated by the first information are equal to a total number of jobs and a total number of pages that are indicated by the second information.

16. The method according to claim 1, wherein the device operation state information indicates a ratio of jobs through the control apparatus and a ratio of jobs processed by the image processing apparatus without going through the control apparatus in a same job type.
